

A working roadworks planning system

Built and running. Real Melbourne data, traffic simulation and AI road-user insights.

REAL DATA FOUNDATION OSM roads | SCATS traffic counts | PTV timetables | City pedestrian counts

PROBLEM STATEMENT

Closures disrupt **drivers, public transport and pedestrians**. Planners need to test the impact, including **overlapping works**, before deployment.

OUR INNOVATION

Four AI driver personas

Commuter / Local / Visitor / Delivery

They read the same sign differently. Their **LLM interpretations** inform the simulation.

AI interprets. The engine calculates.

VALUE & PATH TO MARKET

Help **councils and contractors** compare traffic impact and equipment cost before crews arrive.

Proposed model: RPM Hire pairs impact reports with hire quotes; councils use the evidence for permit review.

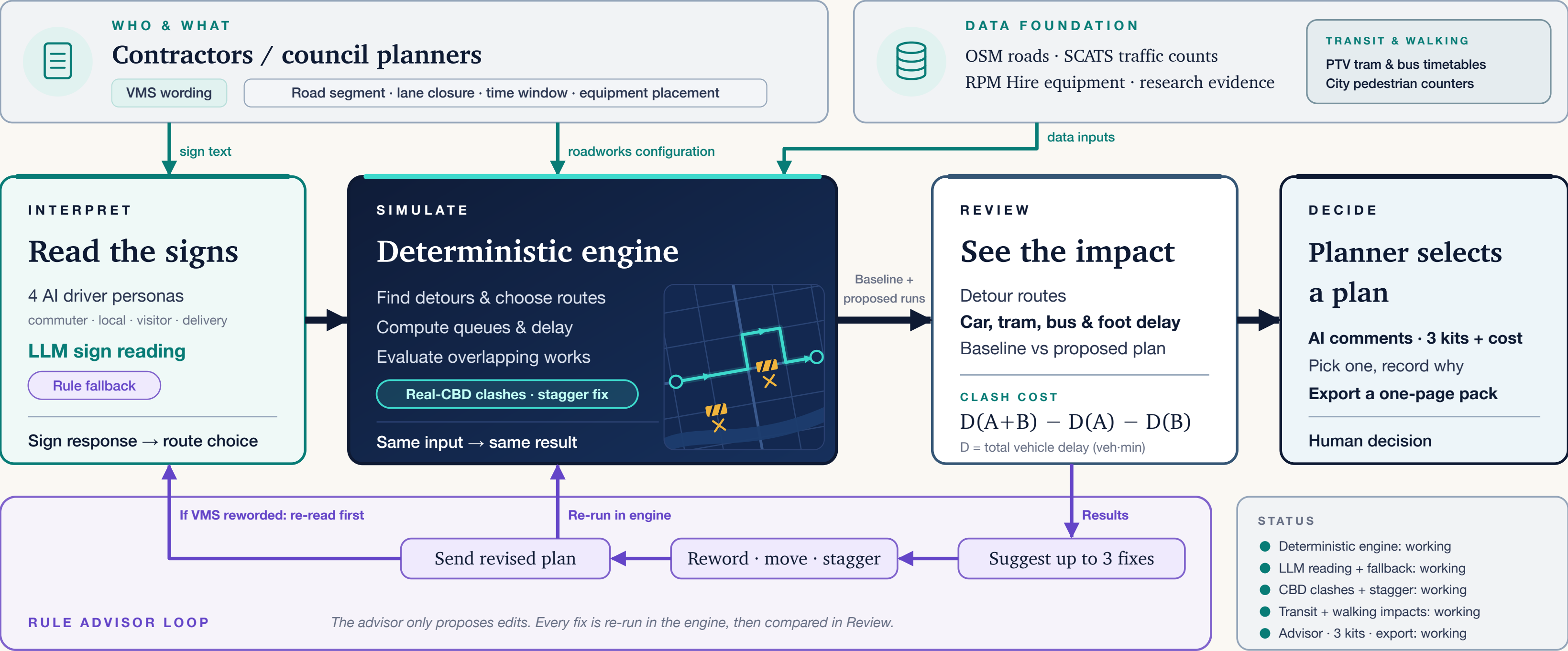
Next: pilot on one worksite.



Live product: cars (blue), pedestrians (purple) and trams (green). La Trobe, 17:00: 0 m queue; Lonsdale, 08:00 with a detour sign: **538 m** (p.3). Movements are illustrative; impacts are engine-calculated.

System architecture

Under the hood: AI interprets. The engine calculates.



BUILT & RUNNING

Four-junction simulation, four AI personas, plan comparison and export.

STACK JavaScript / Canvas · Python / OSMnx

DeepSeek · Cloudflare Workers

NEXT DELIVERY

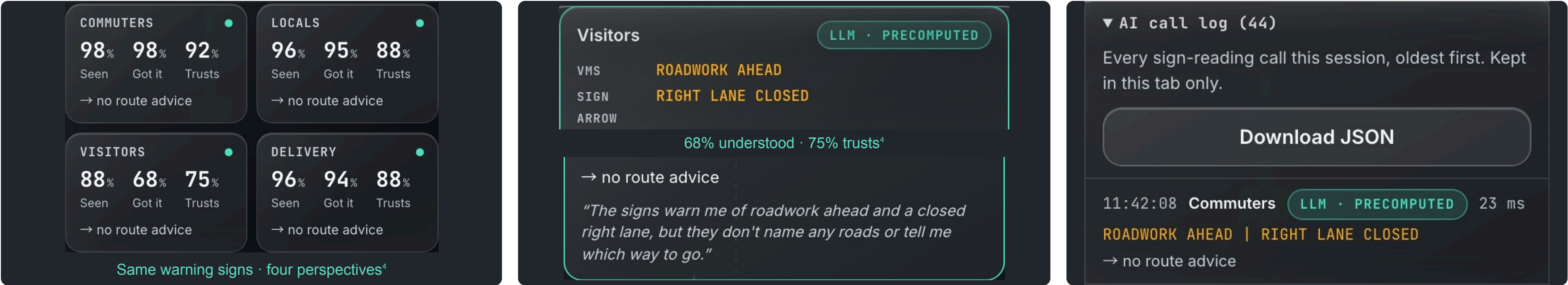
1. Connect the native SUMO backend to the UI.
2. Apply initial weather-backtest findings to model parameters.
3. Validate one worksite case and rehearse the full export flow.

A working system, end to end

Real product screens. From a road closure to a plan you can act on.

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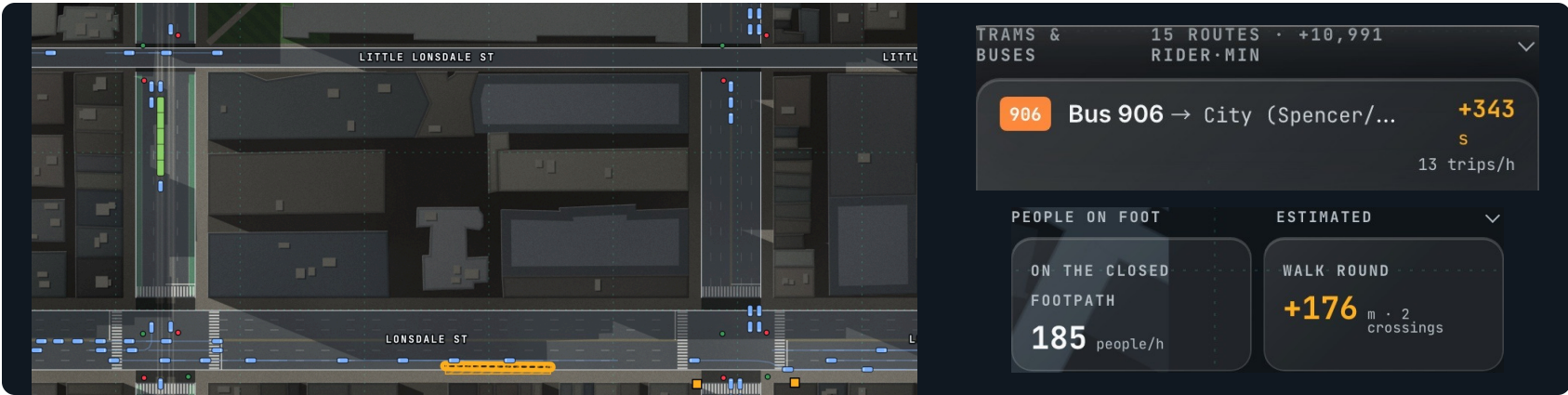
Impact · Four AI personas → full explanations → traceable readings



AI interprets the sign. Each persona explains its reading; the engine calculates route choices and delay.⁴

02–03

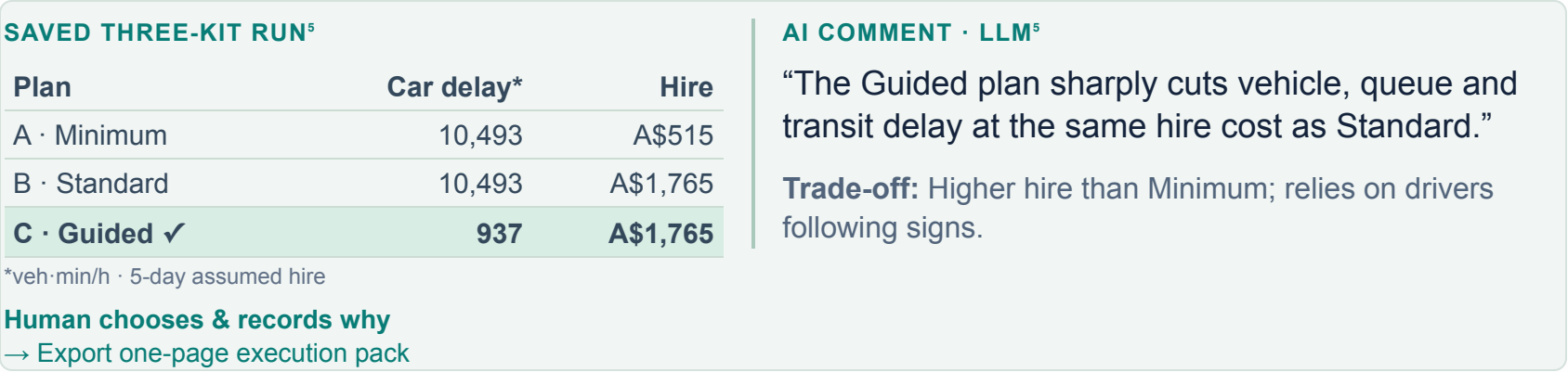
Four junctions · Transit & walking impact



Left: running junction micro-sim. Right: network impact, footpath closed.³

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Improve · AI comments → choose → export



Warning-only equipment adds no predicted diversion; **safety value is not scored**.
C names the detour and the time saved (**USE RUSSELL · SAVE 9 MIN**) on the same A\$1,765 kit.

–46% vehicle delay: add "USE RUSSELL ST"¹

3-day shift ends the overlap between works²

+176 m walk if the south footpath also closes³

MODELLED
SCENARIO RESULTS

¹ Lonsdale WB, weekday 08:00: one lane closed, footpath open. Add "USE RUSSELL ST" to "ROADWORK AHEAD": 10,493 → 5,629 veh·min/h (–46%); 538 m queue. ² Demo register: shift other works by 3 days to end overlap. ³ Same site, south footpath also closed: +176 m; ~185 ped/h, interpolated. ⁴ Warning-only signs: precomputed LLM readings; trust is assumed. Log counts recorded reads, not paid model calls. ⁵ Saved three-kit run, same site/hour, footpath open; LLM comment on those saved outputs. Assumed: passenger loads, driver mix/behaviour, stock and hire rates. Results are modelled, not field measurements.